

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. **100 points:** Part A 14, Part B 16, Part C 40, Part D 30.

Part A — Vocabulary**14 questions, 1 point each**

Set 1 — Data and Sampling. Write the letter of the matching description next to each term.

- | | |
|----------------------------------|---|
| 1. E Categorical variable | A. A number computed from a sample. |
| 2. C Statistical question | B. The population is split into groups that are alike inside, and some are chosen at random from every group. |
| 3. G Parameter | C. A question whose answers are expected to vary, and that names the group and what will be recorded. |
| 4. A Statistic | D. A few whole groups, each a small mix of the whole, are chosen at random and everyone in them is studied. |
| 5. F Sampling frame | E. A variable whose values are labels or group names. |
| 6. B Stratified sample | F. The list of individuals a study can actually choose from. |
| 7. D Cluster sample | G. A number that describes a population. |

Set 2 — Bias and Study Design. Same directions.

- | | |
|--------------------------------------|---|
| 8. D Undercoverage | A. Who asks, the wording, or the order of the questions pushes people away from their true answer. |
| 9. G Nonresponse | B. A variable that differs between an experiment's groups and affects the response, so two explanations cannot be told apart. |
| 10. A Response bias | C. Units are sorted into groups of similar units first, then randomly assigned within each group. |
| 11. F Lurking variable | D. Part of the population is left off the frame and can never be chosen. |
| 12. B Confounding variable | E. A fake treatment built to look real. |
| 13. E Placebo | F. A variable outside the study that affects both variables being compared. |
| 14. C Randomized block design | G. Individuals were chosen for the sample but never answered. |

Part B — Multiple Choice**8 questions, 2 points each**

15. Granite Falls Middle School records each student's **bus route number** (4, 7, 12, or 15). This variable is
- (A) quantitative, because the values are numbers.
 - (B) quantitative, because you can compute a mean route number.
 - (C) categorical, because the numbers are labels for routes. ← **correct**
 - (D) neither, because a route number is not really data.

16. Two volunteers each take an honest random sample of 50 Silver Lake Recreation Center members. One finds that 34% swim laps weekly; the other finds 42%. The best explanation is that
- (A) both are statistics estimating one parameter, and samples vary. ← correct
 - (B) one volunteer made a mistake, because there is only one true percent.
 - (C) the true percent must be exactly 38%, halfway between them.
 - (D) the percent of members who swim changed between the two samples.
17. Each of Maplewood High School's 32 homerooms holds students from a **single grade**. A researcher draws 2 homerooms at random and surveys everyone in them. The main problem with this sample is that
- (A) no chance was used, so it is a convenience sample.
 - (B) two homerooms is too small a sample to use.
 - (C) a cluster sample can never be used in a school.
 - (D) a homeroom is alike inside, so only one or two grades are reached. ← correct
18. Brookside Apartments draws 60 households at random from its complete list and mails each a survey. 22 are never returned. The missing 22 are an example of
- (A) undercoverage, because those households were left out.
 - (B) nonresponse, because they were chosen but never answered. ← correct
 - (C) a convenience sample, because the others were easier to reach.
 - (D) sampling variability, because samples differ from each other.
19. A survey's method causes it to **underestimate** by 12 percentage points. The researcher repeats it with **five times as many people** and the same method. The new estimate will most likely
- (A) underestimate by about the same 12 points. ← correct
 - (B) land close to the true value, because the sample is larger.
 - (C) overestimate, because larger samples pull the other way.
 - (D) be impossible to predict in any direction.
20. In a random sample of adults, those who own a dog walk more minutes per day than those who do not. Nobody was assigned a dog. This result shows
- (A) that owning a dog causes people to walk more.
 - (B) that walking more causes people to buy a dog.
 - (C) an association only; a lurking variable could explain it. ← correct
 - (D) nothing at all, because the sample was chosen at random.
21. In an experiment on a new vitamin, the control group is given a **placebo**. Its purpose is to
- (A) make the total number of subjects larger.
 - (B) let both groups believe they are treated, so only the vitamin differs. ← correct
 - (C) tell each subject which group they are in.
 - (D) remove the need for random assignment.
22. A researcher wants to know whether energy drinks cause heart problems in 12-year-olds. No experiment is run. The best reason is that the experiment would be
- (A) impractical, because 12-year-olds are hard to find.
 - (B) impossible, because heart problems cannot be measured.
 - (C) unnecessary, because observational studies prove causation.
 - (D) unethical, because it would assign children to drink them. ← correct

Part C — Short Answer & Computation

8 questions, 5 points each

23. **Granite Falls Middle School** has 840 students. For each student the office records: bus route number, hours of sleep last night, favorite elective, and number of siblings.
- (a) Who are the **individuals** in this data set? **The 840 Granite Falls students**

(b) Classify each variable.

Variable	Categorical or quantitative?
Bus route number	Categorical
Hours of sleep last night	Quantitative
Favorite elective	Categorical
Number of siblings	Quantitative

(c) The office's first draft question is "*How much sleep?*" Explain why that is not a statistical question, then rewrite it so that it is.

It names no group and no measurement, and a single person has one answer.

"How many hours did Granite Falls students sleep last night?"

24. **Silver Lake Recreation Center** has 1,600 members. The director surveys 80 members chosen at random; 28 of them use the pool every week.

(a) Population: **All 1,600 members** Sample size: **80**

Sample: **The 80 members who were surveyed**

(b) What percent of the sample uses the pool weekly? About how many of all 1,600 members does that estimate?

$$\frac{28}{80} = 0.35 = 35\%$$

$$0.35 \times 1600 = 560 \text{ members}$$

(c) Is the 35% a parameter or a statistic? Say how you know.

A statistic — it describes the 80-member sample, not all 1,600 members.

(d) Name one **constraint** that keeps the director from taking a census of all 1,600 members.

Time or cost — reaching every one of 1,600 members would take far too long.

25. **Maplewood High School** has 960 students and wants a **stratified** random sample of 64, allocated in proportion to grade size.

(a) What fraction of the school is being sampled?

$$\frac{64}{960} = \frac{1}{15}$$

(b) Show the work for the **10th grade** row, then complete the table.

$$255 \times \frac{1}{15} = 17 \text{ students}$$

Grade	9	10	11	12	Total
Students in grade	300	255	225	180	960
Number sampled	20	17	15	12	64

26. **Brookside Apartments** has 720 households on a numbered list. The manager wants a **systematic** random sample of 48.

(a) Find the interval k .

$$k = 720 \div 48$$

$$k = 15$$

(b) The random start is household 7. List the first four households selected:

7 22 37 52

(c) Name the sampling technique each manager used.

What the manager did	Technique
Split the households by building age and drew some at random from every building, in proportion.	Stratified
Drew 4 of the 24 floors at random, then surveyed every household on those 4.	Cluster
Stood in the lobby on Saturday morning and asked whoever walked past.	Convenience

27. **Elmhurst Public Library** has 1,500 card holders and wants to know what fraction want Sunday hours. Two random phone samples were taken.

Method 1. 100 card holders were asked, “*Most libraries now open on Sundays. Wouldn’t you like Sunday hours too?*” 74 said yes.

Method 2. 120 card holders were asked, “*Should the library add Sunday hours?*” 54 said yes.

(a) Find the percent saying yes for each method.

$$\frac{74}{100} = 0.74 = 74\%$$

$$\frac{54}{120} = 0.45 = 45\%$$

(b) Estimate the number of all 1,500 card holders each method points to.

$$0.74 \times 1500 = 1110 \text{ card holders}$$

$$0.45 \times 1500 = 675 \text{ card holders}$$

(c) Both samples were random, yet one method is biased. Name it, name the **type** of bias, and say which **direction** it pushes the estimate.

Method 1. Response bias — the leading wording pushes people toward yes, so it overestimates: 74% against 45%, or 1,110 card holders against 675.

28. A random sample of 200 Maplewood students is taken. Of the 80 who play a **school sport**, 60 made the honor roll. Of the 120 who do not, 60 made the honor roll. Nobody was assigned — students chose.

(a) Find each group’s honor-roll percent, and the gap between them.

$$\frac{60}{80} = 0.75 = 75\%$$

$$\frac{60}{120} = 0.50 = 50\%$$

$$75\% - 50\% = 25 \text{ percentage points}$$

(b) Explanatory variable: **Plays a school sport** Response: **Made the honor roll**

(c) Using all 200 students, find the overall honor-roll percent and estimate how many of all 960 students that is.

$$\frac{60 + 60}{200} = \frac{120}{200} = 0.60 = 60\%$$

$$0.60 \times 960 = 576 \text{ students}$$

(d) Maplewood requires passing grades to stay on a team. Name the **reversed arrow** or a **lurking variable**, and say which way it pushes the numbers.

Reversed arrow: good grades are what let a student stay on a team, so grades put students into the sport group — that pushes the athletes’ percent up on its own.

29. Maplewood now runs an **experiment**. Of 80 volunteers, a random number generator sends 40 to a review session before the unit test and 40 to an ordinary study hall. Afterwards 30 of the review group and 22 of the study hall group passed.
- (a) Find each group's pass percent and the gap.

$$\frac{30}{40} = 0.75 = 75\%$$

$$\frac{22}{40} = 0.55 = 55\%$$

$$75\% - 55\% = 20 \text{ percentage points}$$

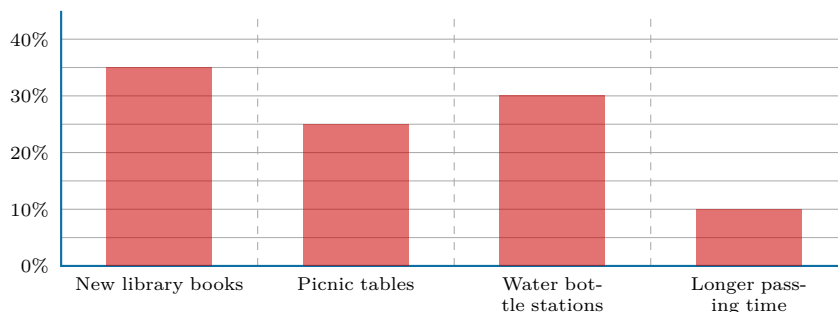
- (b) Treatment: **Review session** Control group: **The 40 in study hall**
- (c) Instead, Maplewood **blocks** on the last unit test: 48 volunteers passed it and 32 did not. Randomly split each block in half. How many go to each group?

$$48 \div 2 = 24 \text{ per group (passed block)}$$

$$32 \div 2 = 16 \text{ per group (did-not-pass block)}$$

30. The Maplewood Student Council asked a stratified sample of 80 students how to spend a grant. Complete the table, then complete the bar graph.

Choice	Count	Relative frequency
New library books	28	35%
Picnic tables	20	25%
Water bottle stations	24	30%
Longer passing time	8	10%



The council's first draft item read: "Don't you agree the grant should buy new books and picnic tables?" Name the **two** instrument rules it breaks.

Leading — "Don't you agree" pushes toward yes, so the wording is not neutral.

Double-barrelled — it asks about books and picnic tables in one item.

Part D — Extended Response

3 questions, 10 points each

31. Elmhurst Public Library has run **two** studies of its summer reading program.
- Study A.** Library records were checked. Of the children whose families *chose* to sign up, 84% met their reading goal; of those who did not sign up, 52% did.
- Study B.** Of 120 volunteer children, a generator *assigned* 60 to the program and 60 to no program. Librarians collected the data by *observing* each child read aloud: 75% of the program group met the goal and 60% of the others did.
- (a) Which study is observational and which is an experiment? Give the one feature that decides it — and say why Study B's *observing* does not change your answer.
- A is observational, B is an experiment: who decided the groups — families chose in A, a generator assigned in B. Observing is how B collected data, not how it built groups.**
- (b) Study A's gap is 32 points and Study B's is 15. Which study lets the library write "the program caused more children to meet their goal"? Justify it.

Study B. Random assignment made the groups alike at the start, so the program is the only difference left. Study A's bigger gap earns only "is associated with."

(c) Explain why Study A's gap came out so much *bigger* than Study B's.

Families who chose to sign up were already readers, so A's program group started ahead. Self-selection inflates the gap; in B nobody chose, so that head start is gone.

32. **Brookside Apartments** mailed a survey about a proposed dog park to all 720 households. 180 were returned, and 126 of those said yes. The management office's own sign-up records show that 288 of the 720 households actually want one.

(a) Find the response rate, the survey's percent saying yes, and the number of households that percent points to.

$$\frac{180}{720} = 0.25 = 25\% \text{ response rate}$$

$$\frac{126}{180} = 0.70 = 70\%$$

$$0.70 \times 720 = 504 \text{ households}$$

(b) The records say the truth is 288 of 720, or 40%. Name the bias, and give its size in percentage points *and* in households.

Nonresponse bias — the 75% who never returned it cared less about a dog park.

It overestimates by 30 percentage points, or $504 - 288 = 216$ households.

(c) The manager mails again and gets 360 returns, 252 of them yes. Compute that percent and explain what the result shows.

$$\frac{252}{360} = 0.70 = 70\%$$

Twice the responses, the same wrong 70%: a larger sample does not remove bias.

(d) Describe one change that would actually reduce the bias, and name the bias it removes.

Draw a random sample of households and follow up by phone or in person until nearly all of them answer. That removes the nonresponse bias, which size never did.

33. **Granite Falls Middle School** (840 students) wants to know whether to start a school garden club. The office has a numbered list of all 840 students, and the three grades are believed to differ a lot in their answers.

(a) Choose **one** of the four sampling techniques and justify it from what this context allows.

Stratified. A full numbered list exists and the grades are expected to differ, so drawing at random within each grade guarantees all three are represented in proportion.

(b) Name one of the five **data collection methods** and say why it fits this question.

Survey. The question has a fixed yes/no answer and needs many students, so a short written instrument given to the whole sample can be counted quickly.

(c) Instead, a teacher asks her own advisory of 30 students; 12 say yes. A random sample of the school later finds 38%. She reports: "My advisory got 40% — within 2 points — so asking one advisory works as well as a random sample." Explain what is wrong with her claim, and state what she *is* allowed to say.

Chance never chose her advisory — the other 810 students could not be asked — so the match is luck: the same room could have missed by 20 points and nothing would warn her. She may say only that 40% of her own advisory wants a garden club.